SEG3125 Assignment 5 - Bilingual Interactive Dashboard

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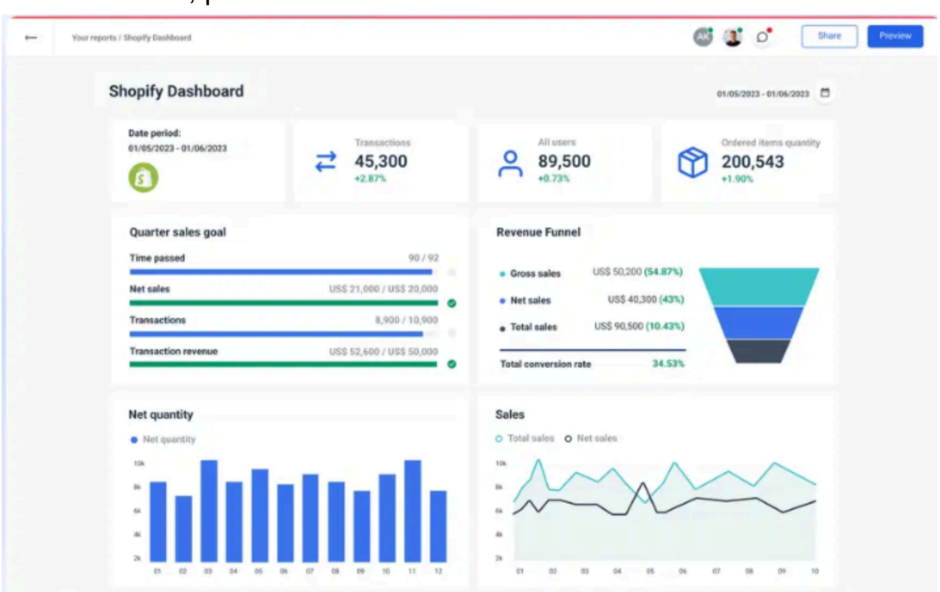
1. Dashboard Goal + Data

Goal & Domain:
The dashboard, titled "Mainstream Music Evolution," visualizes the cultural shifts in popular music from the 1960s to the 2020s. It focuses on the music domain, specifically analyzing genre market shares and top-performing songs on the Billboard Hot 100.

Dataset & Modifications:
The dataset draws from some real historical data but is ultimately synthetic, generated using AI to model historical Billboard Hot 100 trends realistically. It tracks six major genres (Pop, Rock, Hip-Hop, R&B, Country, Electronic) across seven decades. As per the assignment requirements, a clear disclaimer is included in the dashboard footer stating that the data is synthetic and for educational purposes.

Inspiration Dashboards:
The structural flow was inspired by the Whatagraph Shopify dashboard (https://whatagraph.com/shopify-dashboard), which excels at presenting complex e-commerce data in a clean, professional interface.



Visually, however, the dashboard draws heavy inspiration from 1920s Bauhaus posters (e.g., works by Joost Schmidt and Herbert Bayer), translating their strict constructivist principles of primary colours, thick borders, and geometric grid layouts into a functional data visualization interface.

2. Reflection/Design

(A) Charts

I decided on two distinct chart types to represent the data from different analytical perspectives:

- **Genre Trend Chart (Line / Area / Stacked Area):** This chart visualizes the estimated market share (%) of each genre over the decades. A time-series chart (line/area) was chosen because it is the most effective way to show continuous change over time and compare the relative trajectories of multiple categories (genres).
- **Top Songs Chart (Horizontal Bar Chart):** This chart displays the top songs ranked by their peak weeks at #1 or total chart weeks. A horizontal bar chart was selected because it allows for easy visual comparison of discrete items (songs) while accommodating long song titles cleanly along the Y-axis without text rotation or overlap.

(B) 3Cs (Context, Clutter-Free, Contrast)

- **Context:** The dashboard immediately introduces its purpose with a clear title ("Mainstream Music Evolution") and a brief introductory paragraph explaining how to use the interactive controls. Furthermore, each chart includes a descriptive subtitle explaining exactly what the data represents.
- **Clutter-Free:** Embracing the Bauhaus ethos of "form follows function," the UI is stripped of unnecessary ornamentation. I removed superfluous gridlines, gradients, and drop shadows from the charts. The layout relies on generous negative space and stark geometric divisions to separate sections logically, preventing cognitive overload.
- **Contrast:** The visual hierarchy is driven by extreme contrast. I used a strict palette of stark primary colours (Red #D02020, Blue #1040C0, Yellow #F0C020) set against solid black and white backgrounds. To ensure clarity across the 6 genres, each is assigned a unique, distinct colour that remains easily distinguishable within the dashboard. This approach not only fulfills the thematic aesthetic but also ensures excellent readability and instantly draws the user's eye to key data points, active filters, and interactive elements.

(C) Layout, Title, and Interactions

- **Layout & Title:** The dashboard is a single-page application titled "Mainstream Music Evolution." It features a top-down flow: Header & Intro -> Global Filters & Stats -> Charts.
- **Interactions:** The user has extensive control over the data being displayed:

    - **Decade Selector:** Users can filter the entire dashboard to view a specific decade (e.g., 1980s) or view all historical data simultaneously.
    - **Genre Filtering:** Users can toggle specific genres on and off to isolate trends (e.g., comparing just Rock vs. Hip-Hop).
    - **Chart Toggles:** The Trend chart allows users to swap between Line, Area, and Stacked views. The Bar chart allows sorting by "Weeks at #1," "Chart Weeks," "Chronologically," or alphabetically.

(D) Internationalization

- **Languages Chosen:** English (Default) and French.
- **Translation Source:** The translations were obtained mostly using Google Translate and GenAI to ensure contextually accurate phrasing for UI elements.
- **Translation Difficulties:** Direct translation often resulted in text expansion. For example, "Chart Weeks" translates to "Semaines au Classement," which was significantly longer. I had to ensure the CSS flexbox layouts and chart margins were responsive enough so that the longer French strings did not break the UI or overlap with other elements.
- **Localization Process:** I implemented a centralized dictionary (translations.js) and localized all textual elements. This includes the global headers, introductory text, axis labels, chart tooltips, legends, interactive button labels, dynamic insight sentences, and footer disclaimers.
- **Beyond Text: Localizing Non-Textual Elements:** Beyond UI strings, I localized non-textual elements to address identified gaps. I used *Intl.NumberFormat* for region-specific number conventions (e.g., decimal separators) and rendered years as plain strings to prevent incorrect grouping separators (e.g., "2024" instead of "2 024"). Decade labels were mapped to French equivalents (e.g., "Années 60"), and the *<html lang>* attribute updates dynamically on language switches to ensure proper accessibility.

3. High-Fidelity Prototype

(A) Visual Design Choices

The visual design of the "Mainstream Music Evolution" dashboard was meticulously crafted to reflect the bold, geometric spirit of the 1920s Bauhaus movement, while ensuring the complex musical data remained accessible and clean.

- **Colour Theme and Consistency:** I established a strict Bauhaus-inspired palette using primary colours: "Red" (#D02020), "Blue" (#1040C0), and "Yellow" (#F0C020), set against solid black and white backgrounds. To ensure the six musical genres (Pop, Rock, Hip-Hop, R&B, Country, Electronic) remain easily distinguishable, each genre is assigned a unique, vibrant colour that stands out clearly against the primary interface colours, ensuring the user can instantly identify data series despite the surrounding primary colour scheme.

- **Typography:** I employed a bold, sans-serif typeface to align with the Bauhaus commitment to functional, geometric clarity. This choice prioritizes legibility for chart labels, axes, and interactive elements, ensuring the technical data is never obscured by decorative fonts.
- **Layout and Negative Space:** The interface leverages the "less is more" principle, utilizing generous negative space to prevent the complex historical data from overwhelming the user. The layout follows a rigid constructivist grid, separating the intro, global filters, and individual charts into clear, geometric zones that provide a logical, top-down narrative flow.
- **Contrast, Scale, Balance, and Visual Hierarchy:**
    - **Contrast:** Critical interactive elements and headers utilize the high-contrast Red and Blue primary tones, ensuring they pop against the white dashboard background.
    - **Scale and Hierarchy:** The dashboard employs a clear hierarchy: the main title is the largest element, followed by descriptive chart subtitles, and finally, the precise data labels within the charts.
    - **Balance:** Asymmetrical balance is used to separate the fixed global controls on the left from the dynamic data visualization areas on the right, maintaining a sense of order despite the high volume of information.
- **Gestalt Principles for Perception:**
    - **Similarity:** All genre-related buttons and filter toggles share consistent hover states and colors, reinforcing their interactive nature.
    - **Proximity & Enclosure:** Charts are contained within clearly defined boundaries with consistent margins, grouping titles, legends, and the data visualization itself as a single cognitive unit.
    - **Continuity:** The layout guides the eye naturally through a logical sequence: moving from the explanatory intro to global filters, and finally down to the interactive charts.
- **Synergy between Visual and Verbal Communication:** The visual aesthetic of the Bauhaus movement serves as a bridge to the historical nature of the data. The stark, constructivist design mimics the radical shifts in musical trends it analyzes, while the clean, rational presentation mirrors the analytical intent of the report, ensuring that the interface feels both historically significant and modernly usable.

(B) Usability Heuristics

To ensure the dashboard is not just visually striking but also highly functional, I incorporated several of Nielsen's 10 Usability Heuristics:

- **Visibility of System Status:** Active filters (e.g., the currently selected decade or active genres) and the current language are clearly highlighted, so users always know exactly what subset of data they are viewing.

- **User Control and Freedom:** Users are never locked into a view. They can easily toggle individual genres on and off or use a "Reset All" function to quickly return the dashboard to its default state.

- **Consistency and Standards:** The colour mapping for genres (e.g., Pop is always Red, Hip-Hop is always Yellow) is strictly maintained across all charts, filters, and tooltips. This internal consistency prevents cognitive friction as the user scans different parts of the dashboard.

4. Code

Portfolio Links:

Live UI (GitHub Pages): https://sai-yarla.github.io/SEG3125-Assignment-1/
Source Code (GitHub Repository): https://github.com/Sai-Yarla/SEG3125-Assignment-1

Dashboard Site Direct Links:

Live UI (GitHub Pages): https://sai-yarla.github.io/SEG3125-Assignment-5/
Source Code (GitHub Repository): https://github.com/Sai-Yarla/SEG3125-Assignment-5

5. Generative AI Acknowledgement

Generative AI played a supportive role in the development of the Mainstream Music Evolution dashboard. AI tools were utilized to generate the synthetic dataset, providing a realistic foundation for the historical Billboard Hot 100 trends. These data points were then manually integrated into the Bauhaus-inspired design system, which I defined through custom typography, grid structures, and a primary colour palette to meet the project's visual objectives.

During the development of the React prototype, GitHub Copilot served as a coding assistant. I leveraged it primarily for scaffolding boilerplate React components, such as initial Recharts configurations, and for troubleshooting complex CSS grid layouts to ensure responsive functionality across screen sizes. Finally, Gemini assisted in structuring this report, ensuring that the content was logically organized and clearly presented in accordance with the assignment requirements.